UNIT42 · FIRMWAREX

Firmware Security Assessment

ARM64 controlled E2E firmware

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Executive summary

This assessment recorded **16 findings** across 2 audited binaries and 6 filesystem entries. Analysis status: **complete**.

Results describe the analysis performed on this image. They do not establish that the firmware is secure or that every candidate is exploitable. Severity and confidence are separate; no CVSS score is inferred.

Critical	High	Medium	Low	Info
0	1	6	9	0

Scope and coverage

Stage	Outcome	Restriction
extract	extracted	See scope limits below
static	completed	See scope limits below
emulation	completed	See scope limits below
symbolic	completed	See scope limits below
secrets crypto	completed	See scope limits below
normalize	completed	See scope limits below

Untrusted execution: blocked_pending_mentor_approval. Raw artifact access: disabled.

- Only registered controlled images execute; other images receive static and scanner analysis.
- Unknown path components are pseudonymized to prevent proprietary-string disclosure.
- Symbolic findings are bounded candidates for human review.

- Only executed stages contribute evidence. Blocked, skipped or incomplete analysis is not a clean bill of health.
- This is an automated firmware-image assessment, not a live-device or network scan.
- Recommendations are general review guidance, not verified fixes.

Binary protection summary

Binary	Architecture	NX	PIE	RELRO	Canary	FORTIFY
/bin/service-probe	ARM64	Present	Not detected	partial	Present	Not detected
/bin/taint-demo	ARM64	Present	Not detected	partial	Not detected	Not detected

Detailed findings

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1. Command injection candidate High

Property	Result
CWE	CWE-78
Affected file	/bin/taint-demo
Architecture	ARM64
Confidence	probable
Review status	new
Address	0x500008
Function / sink	system
Detected	2026-10-20T05:30:38.997302Z

Evidence

Observation	Value
condition satisfiable	true

Observation	Value
Observed taint path	0x4006c4 → 0x400560 → 0x500010 → 0x4006ec → 0x400700 → 0x400550 → 0x500008
Reaching condition (diagnostic expression)	[{"op":"UGE","args":[{"op":"Extract","args":[247,240, {"symbol":"fwx_network_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args":[247,240, {"symbol":"fwx_network_0","bits":248}]}, {"constant":126,"bits":8}]},{"op":"UGE","args": [{"op":"Extract","args":[239,232, {"symbol":"fwx_network_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args … [excerpt; full expression in JSON export]
stack hash	f293294ad2fa580c79354fd807af6a9b7ce710aea9e196297e0d85745846e925

Evidence reference: evidence:e5f46c73d4c68bd594e4e8c63e88995b41c05dc8b2099a7a24544f0a5e5ce950. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review whether external input can reach command execution. Prefer structured APIs and strict allowlists; validate the candidate before remediation.

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2. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/controlled.conf
Architecture	ARM64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.086644Z

Evidence

Observation	Value
rule	credential
fingerprint	1933e5f35126569287a9ff8ae035d160f843c2f48fdb559e992b11c9b9f6bc78
byte offset	9
byte length	19

Evidence reference: evidence:dd3175dc318edd233cc7151c67263b3f8e4ff416f7bc2c366015a1d8a4e46cfd. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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3. Embedded secret candidate Medium

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	ARM64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.086678Z

Evidence

Observation	Value
rule	credential

Observation	Value
fingerprint	61c0a641fa824ceb080db2e491a1d6a3fd31d0c4a9ff9d2f4c6a50fe652c5a12
byte offset	37
byte length	24

Evidence reference: evidence:ee273228aef779ff5c7a067d804ef1f6abda53202b47114bdcf318fada50c4fa. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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4. Embedded secret candidate Medium

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	ARM64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.086699Z

Evidence

Observation	Value
rule	private_key
fingerprint	90e57c8277e852b651a46024f445490d17f66ed6b94578bc65f169088de21c89

Observation	Value
byte offset	62
byte length	78

Evidence reference: evidence:3978394b57ddaa571c27589b566c31720f4f6311073d0e2fc908c1ff9718ea08. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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5. Weak cryptography candidate Medium

Property	Result
CWE	CWE-328
Affected file	/etc/controlled.conf
Architecture	ARM64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.086773Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	a51f23421e28ddbe10be0213589cff4a0b8c5c58e488cc2551315ee4ed2b620d
byte offset	237
byte length	3

Evidence reference: evidence:3c5e77a80b83d1af0675feb2a66e471347a7211fb5282b01e591853103d0867e. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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6. Weak cryptography candidate Medium

Property	Result
CWE	CWE-327
Affected file	/etc/controlled.conf
Architecture	ARM64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.086790Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	94401286a3c175c6d1a8dfaa3afdff19a8a68afdb9c682ee23ce5014d83c022a
byte offset	255
byte length	3

Evidence reference: evidence:e2eea380a1cf2c8e1f59935d3c1ff1cb341a5d28e683d14ec46e1facc330d76c. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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7. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/redacted-fd51334e7491bc1f
Architecture	ARM64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.087032Z

Evidence

Observation	Value
rule	credential
fingerprint	691be9db5767528c706ad0167787688576e59db94b6d9302fe058f5309061481
byte offset	38
byte length	25

Evidence reference: evidence:5ded8f420903f71c556c459a18cd52eed046e9e9d1fffd07d5bed48fa81c09a9. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

8. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/service-probe
Architecture	ARM64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-20T05:30:35.513689Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

9. Embedded secret candidate Low

Property	Result
CWE	CWE-798

Property	Result
Affected file	/bin/service-probe
Architecture	ARM64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.084489Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	e2656a5ba912cc9fe7b6c5c29b022cdef38ddea4f60cc1e1852b98868b2cbd0e
byte offset	369954
byte length	50

Evidence reference: evidence:c540ee12a5151f095c2510103d521dc02ab59fa187d14566fb1017418110b114. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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10. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	ARM64

Property	Result
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.084564Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	8c17ba69cbd86e167838f825cce3ddb3d6e4ea4f36f430a5549c0d9113e26379
byte offset	446993
byte length	26

Evidence reference: evidence:28ea0a72e3cbbe17fc0bdc599f7d110cfc26c7585e4ccd8a1eaaf4078d68b603. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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11. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	ARM64
Confidence	heuristic
Review status	new

Property	Result
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.084602Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	a792ae53d929966addce3173c88ac21e8a8f90b38e0e6757346b61fec4603331
byte offset	447025
byte length	26

Evidence reference: evidence:a584c573b729c19d482448670e14d855fdf841a4f0b4635910314c84c4a0e223. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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12. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	ARM64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable

Property	Result
Detected	2026-10-20T05:30:39.084627Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	3c146d01a2a6effeac6aa38d48561078866d2af677ae4ce35531d46314bd637b
byte offset	447184
byte length	36

Evidence reference: evidence:b0051bf4d2292528a43d172fcc25b46e4d54a4e5920ec9795f34edd3e5ff968f. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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13. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	ARM64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.084650Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	7b951b15c3e4900454eddb7f2f299626919dbff10406d901ab65f5bdca7b4233
byte offset	447232
byte length	36

Evidence reference: evidence:e6a304aee5ca08ad256f6fec6b33c4f95246a6349e735c8d1690cdb701df1784. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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14. Embedded secret candidate Low

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	ARM64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.086717Z

Evidence

Observation	Value
rule	certificate

Observation	Value
fingerprint	db19186572ba25cf0232f46b20e60018e04646c78f18af57efc91c7b6e10245e
byte offset	141
byte length	75

Evidence reference: evidence:966bb39532cf380ac69ba5cf397473c3daa50ce7370632270634b6a239b317ff. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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15. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	ARM64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.086735Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	1fa3068249bcad2db1f2be5c1ec6917ba82fa5d3ed09cbcbd5008d7b2cc94f39

Observation	Value
byte offset	29
byte length	32

Evidence reference: evidence:ea3b751531d9bf95c5ef9e9428588b5b1df135c71bd7eea0341da188a63a7df2. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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16. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	ARM64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:39.086753Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	2d67632135366dc93bc64104fa9b0d54354797a53fb85e2cbb5d4c35645e1174
byte offset	259
byte length	29

Evidence reference: evidence:1e62fb1669f0932226c8aa6fa518ec8e1c7e6d39ec87b96cfbb48d7c5d9615df. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

Assessment identity

Property	Value
Firmware ID	9288256f-696f-4d2c-9a6c-7f2db7e82b2e
SHA-256	acb47f36f2be8b92e7acc0c03f9bb7fde6a28d127c59b1d81782eb96c6168111
Job ID	e4343e5b-d19c-47ee-b5d8-a7d22930c0a4
Image bytes	303104